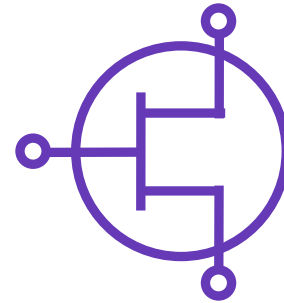


Model Features

- Broadband (DC to 26GHz)
- Large-signal model (Angelov-based)
- Bias scalable: VDSQ (12V to 28V)
- Temperature scalable: (25C to 85C)
- Advanced model feature: enabling intrinsic I-V sensing
- Measurement Validations:
 - Pulsed I-V (25C to 85C)
 - Multi-biased S-parameters (25C to 85C)
 - Single Tone Power and Load Pull (3 & 10 GHz)



HMT-QOR-TGF2023-2-02-001
Qorvo TGF2023-2-02
GaN on SiC HEMT

This model was last revised in Modelithics' Qorvo GaN Library ver. 1.6.5. This is a beta release with tuning parameters for Qorvo's Internal Use, model was refit to improve 10GHz PAE. Previous version released 1.4.9 is available under HMT_TQT_TGF2023_2_02_001_MDLXTQTGaN

Model Description

The HMT-QOR-TGF2023-2-02-001 is a non-linear model for the Qorvo TGF2023-2-02, a discrete 2.5 mm GaN on SiC HEMT using Qorvo's TQGaN25 process. The model is based on the extraction of a customized Angelov non-linear model that is validated against I-V, S-parameters & large signal load pull measurement datasets. The model is extracted such that performance can be scaled with temperature and quiescent bias voltage.

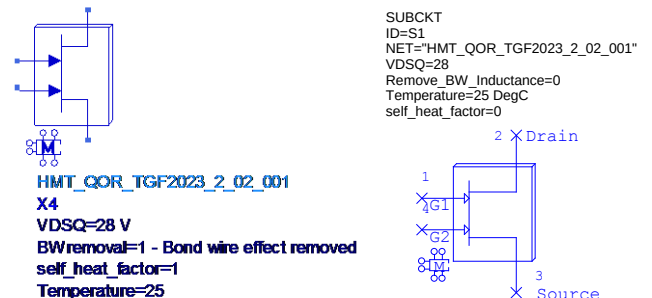
Technical Notes

- Model is optimized for 12-28V operation.
- Model Parameters:
 - **VDSQ**: for setting the optimum bias point of the model.
 - **Temperature**: represents the backside ambient temperature.
 - **Self_heat_factor**: scaling factor for the electrothermal model (range from 0 to 1), 0= self-heating is turned off, 1 (default)= self-heating is fully turned on, and a value of 0.1 is representative of 10% thermal duty cycle.
 - **BWremoval**: 0 includes test fixture used in measurements, 1 (default) sets model reference planes at the center of the gate/drain bond pad (s) and includes source vias effects.

Model Simulation Settings

- **DC I-V**: BWremoval: 0, self_heat_factor: 0 for negligible power dissipation; Temperature: 25C
- **S-Parameters**: BWremoval: 0, self_heat_factor: 1 for CW bias, Temperature: 25C and 85C
- **Load Pull Validations and Single-tone Power Sweeps**: BWremoval: 0, self_heat_factor: 0.1 for 10% thermal duty cycle; Temperature: 25C.

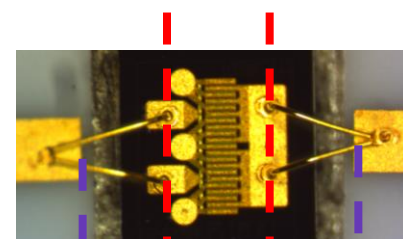
Model Representation



Keysight ADS

NI AWR

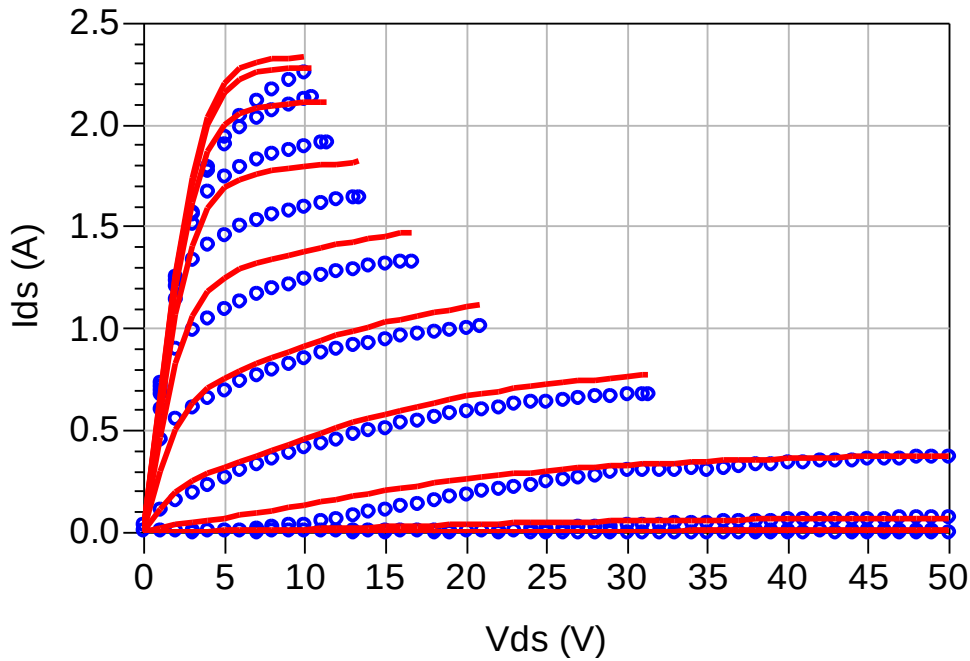
Reference Planes



Model and Model and Measurement Reference Planes (BWremoval=1)

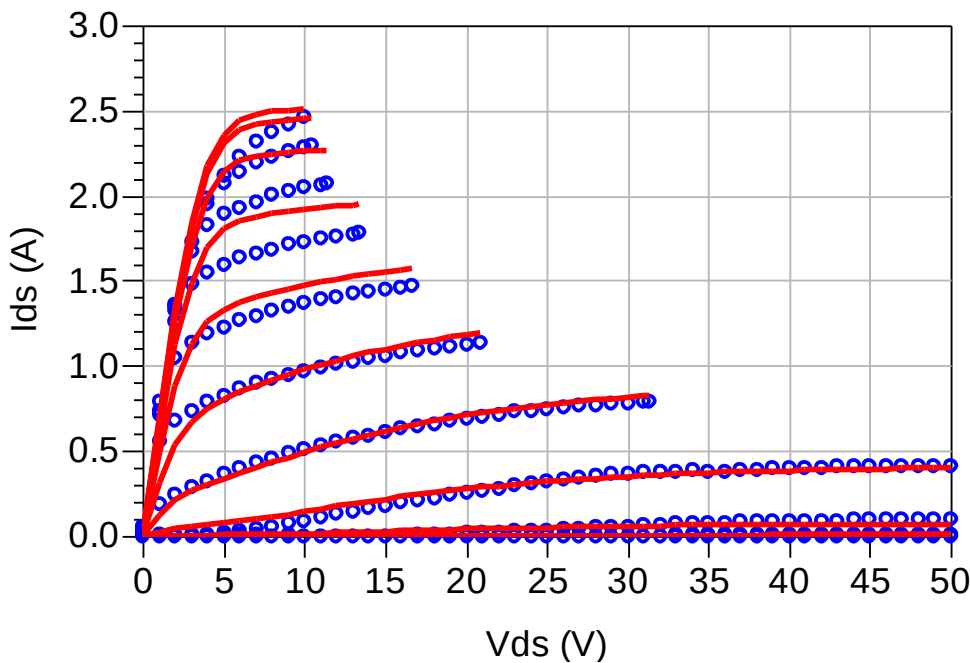
Model and Model and Measurement Reference Planes (BWremoval=0)

DC I-V Characteristics: VDSQ = 28V, 25C



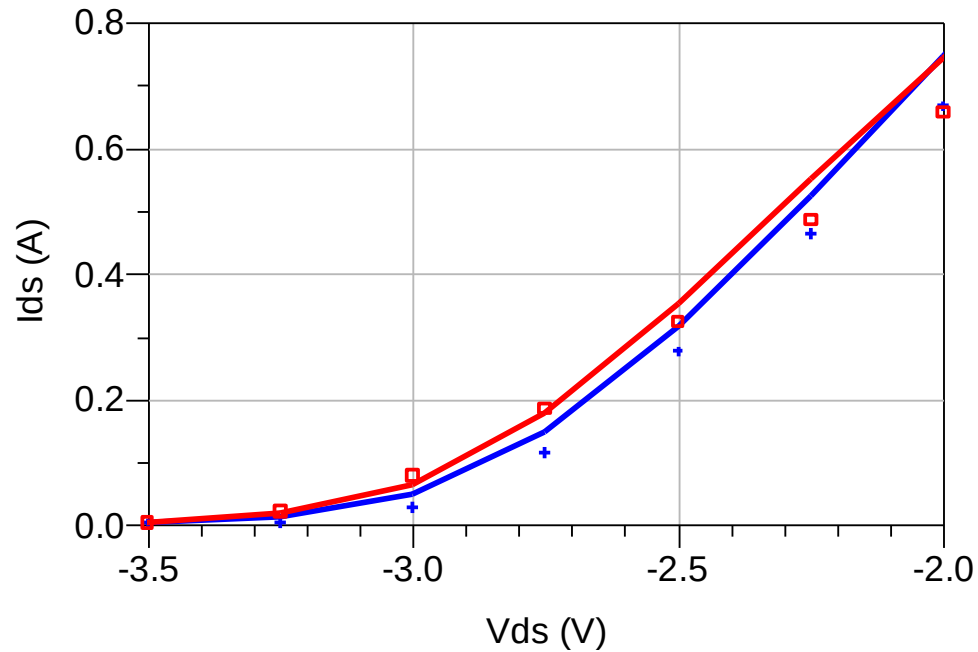
Legend: Red Solid lines - Model data, ○ Symbols - Measured data
 Simulated at 25C with VGS varying from -4 to 1V in steps of 0.5V,
 VDS varying from 0 to 50V in steps of 1V. Model self_heat_factor = 0 and VDSQ = 28V.

DC I-V Characteristics: VDSQ = 12V, 25C



Legend: Red Solid lines - Model data, ○ Symbols - Measured data
 Simulated at 25C with VGS varying from -4 to 1V in steps of 0.5V,
 VDS varying from 0 to 50V in steps of 1V. Model self_heat_factor = 0 and VDSQ = 28V.

Model vs. Measurement Temperature IV Characteristics, VDSQ = 28V

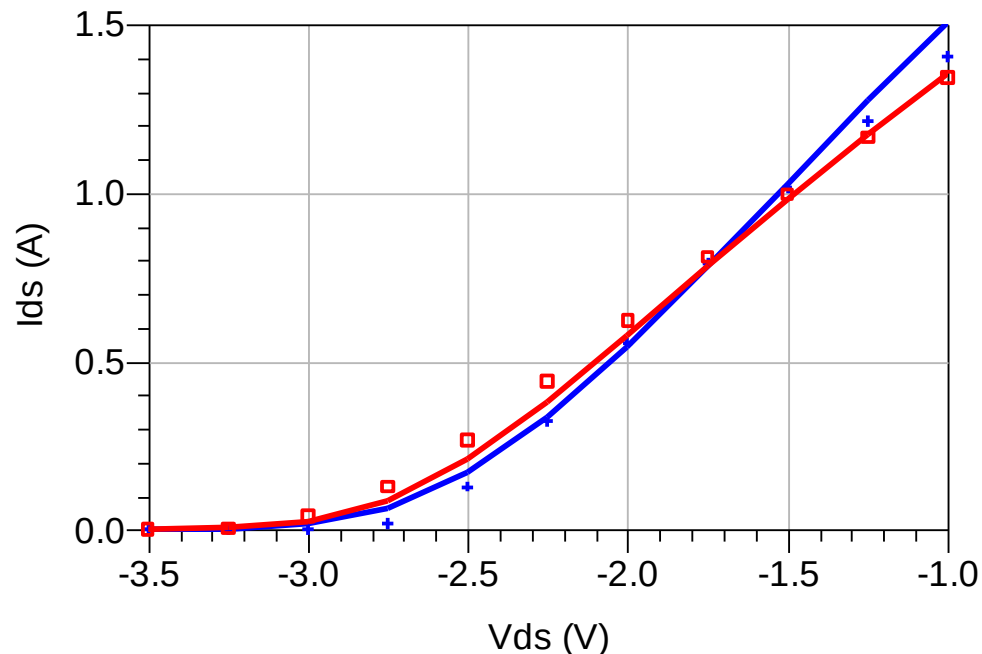


Legend: Red Solid lines: 85C, Blue Solid lines: 25C.

Solid lines - Model data, Symbols - Measured data

Simulated at 25C and 85C with VGS sweep, VDS of 28V. Model self_heat_factor = 0.

Model vs. Measurement Temperature IV Characteristics, VDSQ = 12V

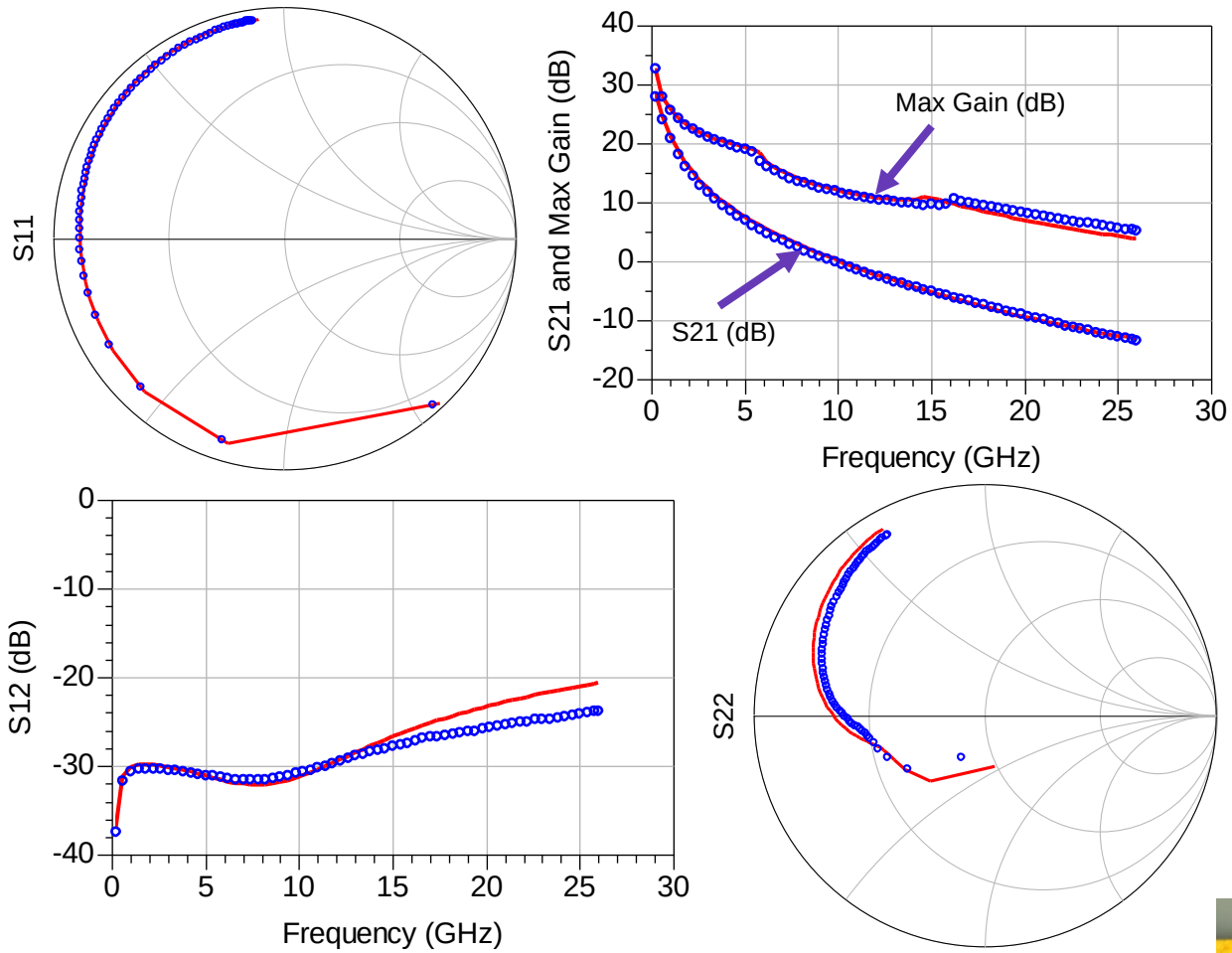


Legend: Red Solid lines: 85C, Blue Solid lines: 25C.

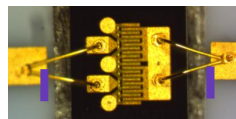
Solid lines - Model data, Symbols - Measured data

Simulated at 25C and 85C with VGS sweep, VDS of 28V. Model self_heat_factor = 0.

S-Parameters Model vs. Measured: $V_{DS} = 12V$, $I_{DS} = 125mA$, $25C$

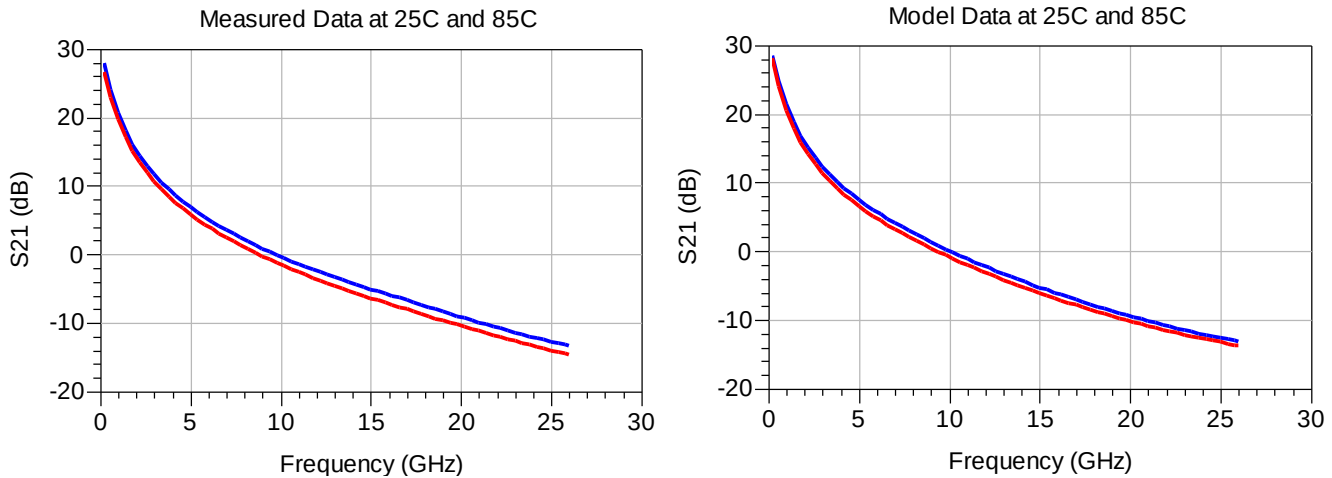


Legend: Red Solid lines - Model data, O Symbols - Measured data
Simulated at 25C with the frequency range from 0.2 - 26GHz. 50Ω Smith Charts



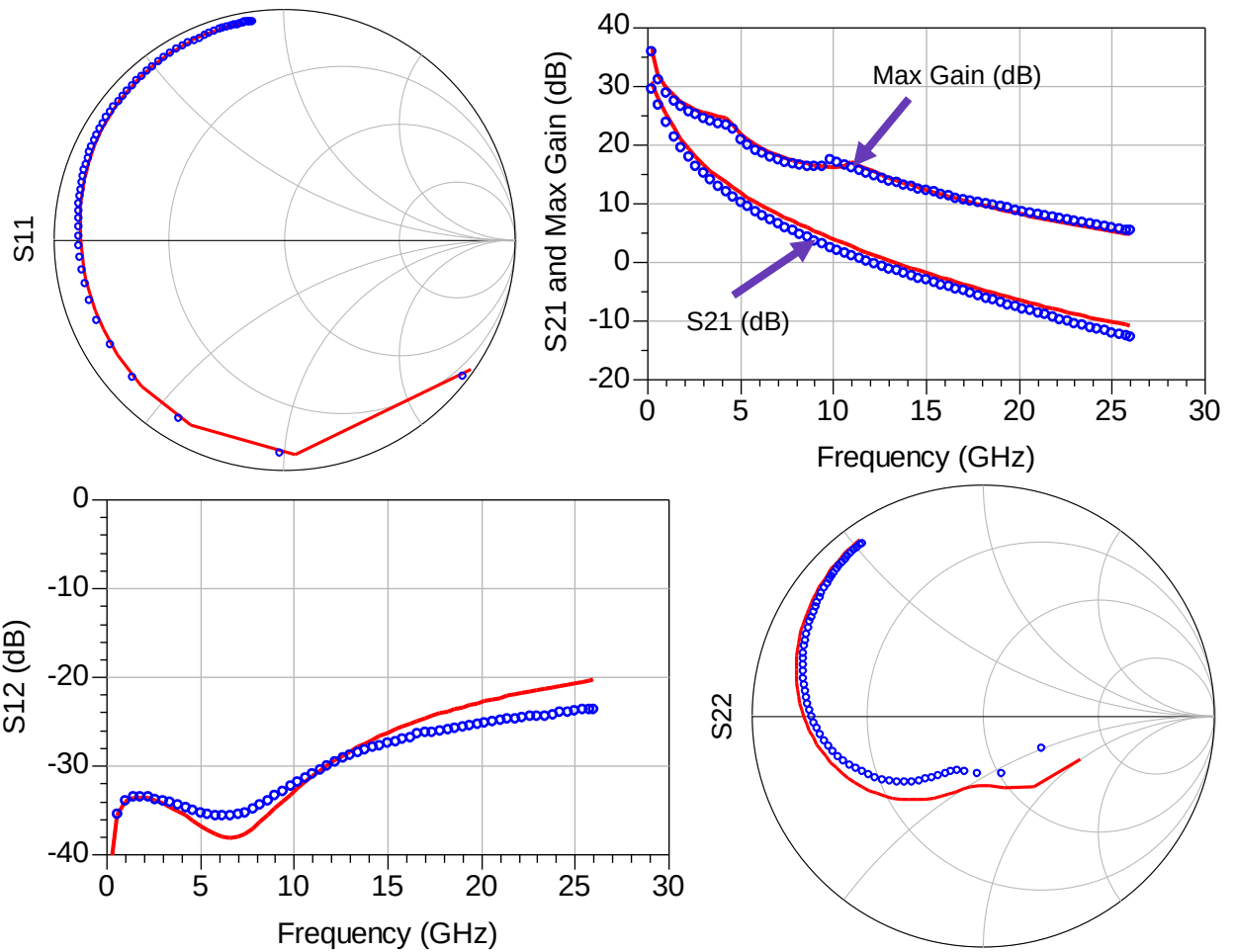
Model and Model
and Measurement
Reference Planes
(BWremoval=0)

S-Parameters vs. Temperature: $V_{DS} = 12V$, $I_{DS} = 125mA$, $25C$



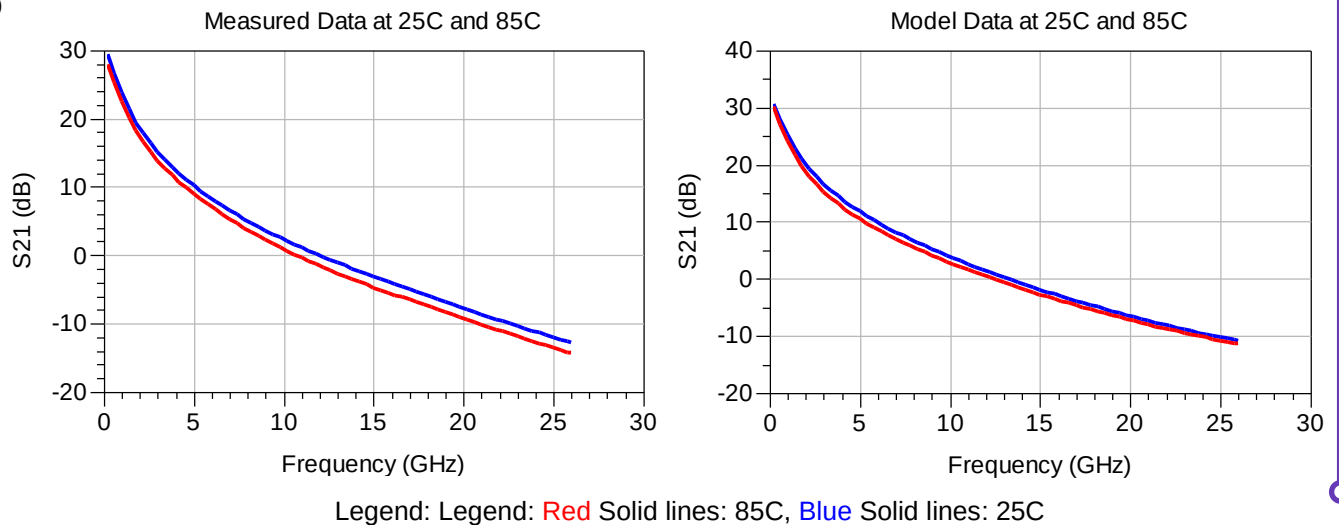
Legend: Legend: Red Solid lines: 85C, Blue Solid lines: 25C

S-Parameters Model vs. Measured: $V_{DS} = 28V$, $I_{DS} = 125mA$, $25C$



Legend: Red Solid lines - Model data, O Symbols - Measured data
Simulated at 25C with the frequency range from 0.2 - 26GHz. 50Ω Smith Charts

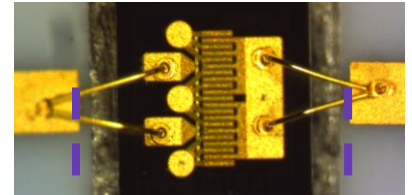
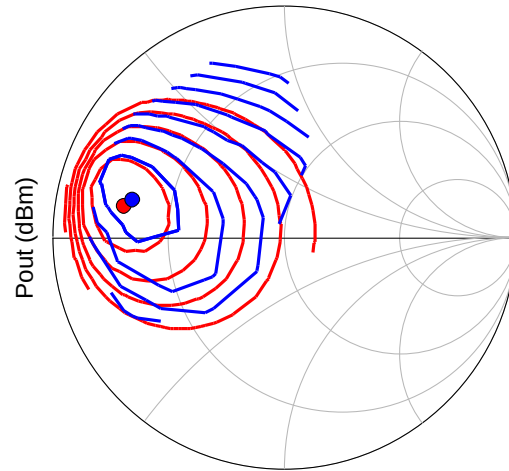
S-Parameters vs. Temperature: $V_{DS} = 28V$, $I_{DS} = 125mA$, $25C$



Load Pull Validation:

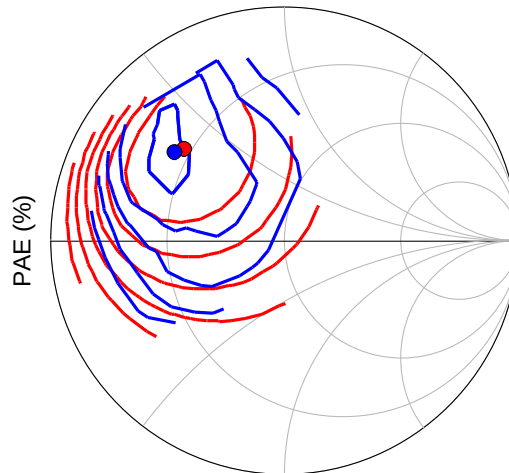
Frequency = 3GHz, VDS = 12V, IDS = 125mA,
Input Power = 18dBm, Z0 = 50Ω Center, 25C

Power Tuning (1dB contour step)



Model and Model and
Measurement Reference
Planes (BWremoval=0)

Efficiency Tuning (10% contour step)



Test Bench Reflection
Coefficient:

GammaS=0.836<163°
GammaS2=0.472<75°
GammaS3=0.729<134°
GammaL2=0.803<54°
GammaL3=0.734<-18°

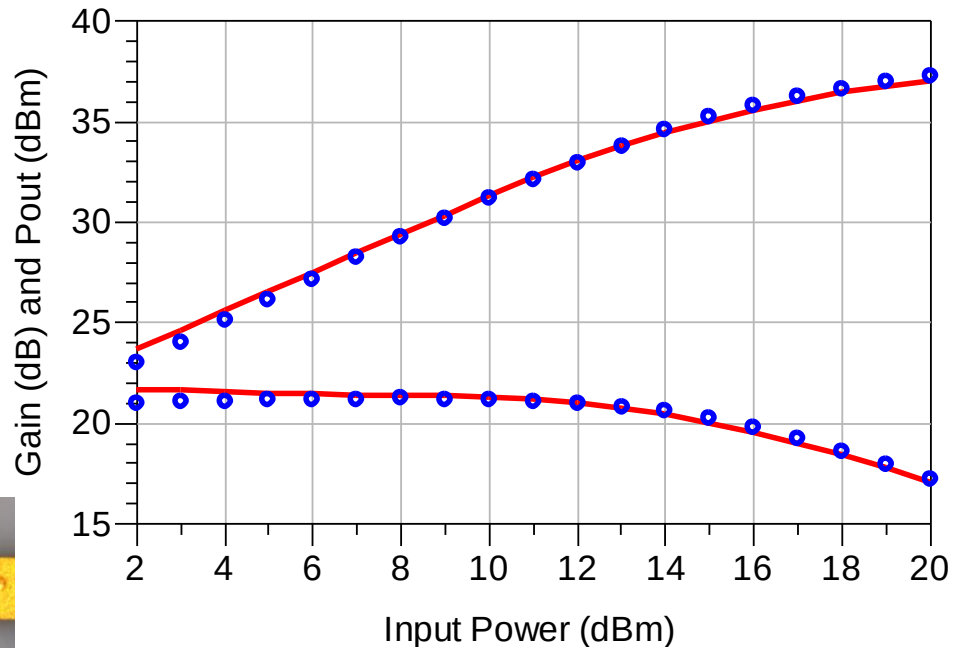
Legend: Red Solid lines – Model, Blue Solid lines – Measured.

Pulsed Biased conditions for measurements: duty cycle = 10% with pulse length = 100us.

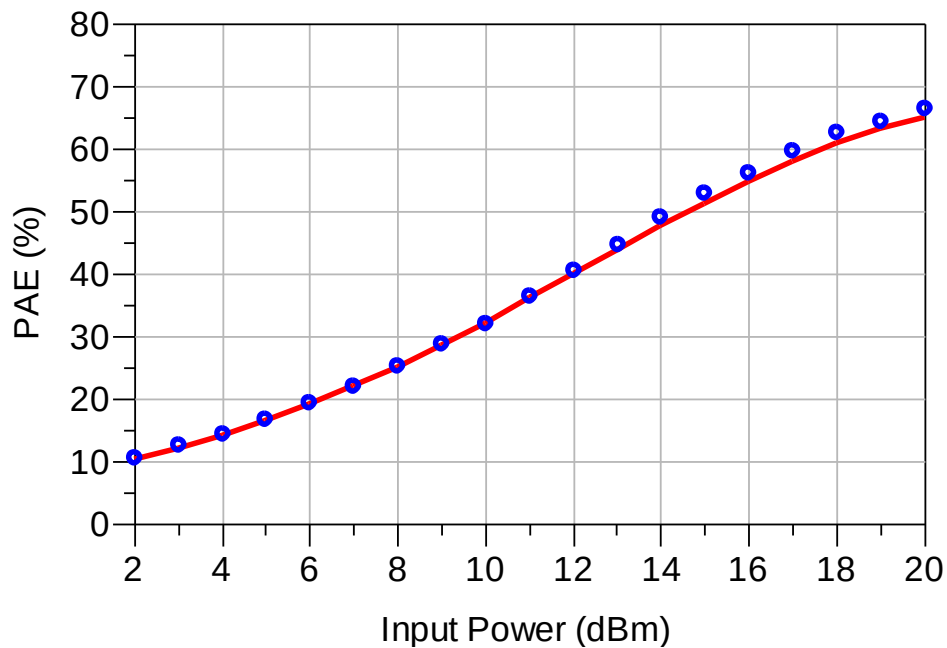
Load Pull Summary	Max Power Load Reflection Coefficient	Max Power Value (dBm)	Max PAE Load Reflection Coefficient	Max PAE Value (%)
Measured	0.735<160°	36.7	0.581<144°	75.2
Model	0.709<169°	36.7	0.576<139°	76.2

Single Tone Power Sweep: Frequency = 3GHz
VDS = 12V, IDS = 125mA, 25C
Load Condition: Power Tuned

Transducer Gain and Output Power

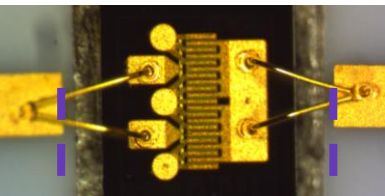


Power Added Efficiency (PAE)



Legend: Red Solid lines - Model data, O Symbols - Measured data

Pulsed Biased conditions for measurements: duty cycle = 10% with pulse length = 100us.



Model and Model and
Measurement Reference
Planes (BWremoval=0)

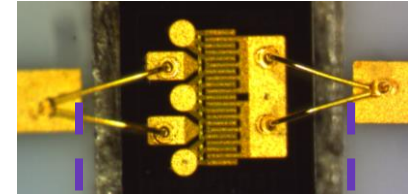
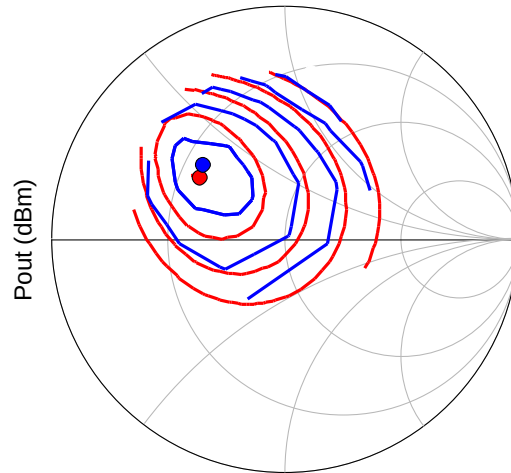
Test Bench Reflection
Coefficient:

GammaS=0.836<163°
GammaS2=0.472<75°
GammaS3=0.729<134°
GammaL=0.679<166°
GammaL2=0.742<100°
GammaL3=0.676<42°

Load Pull Validation:

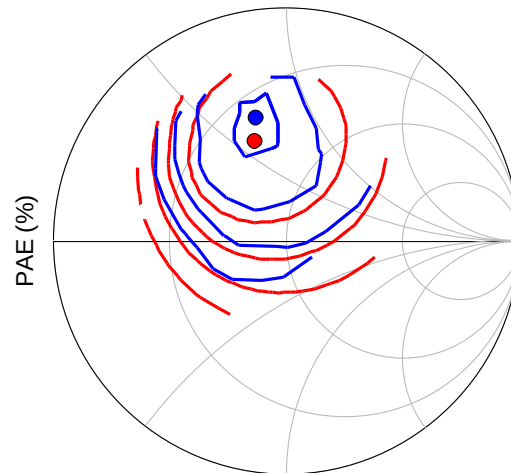
Frequency = 3GHz, VDS = 28V, IDS = 50mA,
Input Power = 19dBm, Z0 = 50Ω Center, 25C

Power Tuning (1dB contour step)



Model and Model and
Measurement Reference
Planes (BWremoval=0)

Efficiency Tuning (10% contour step)



Test Bench Reflection
Coefficient:

GammaS=0.817<163°
GammaS2=0.419<74°
GammaS3=0.662<132°
GammaL2=0.314<-32°
GammaL3=0.407<-170°

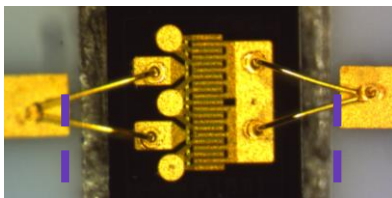
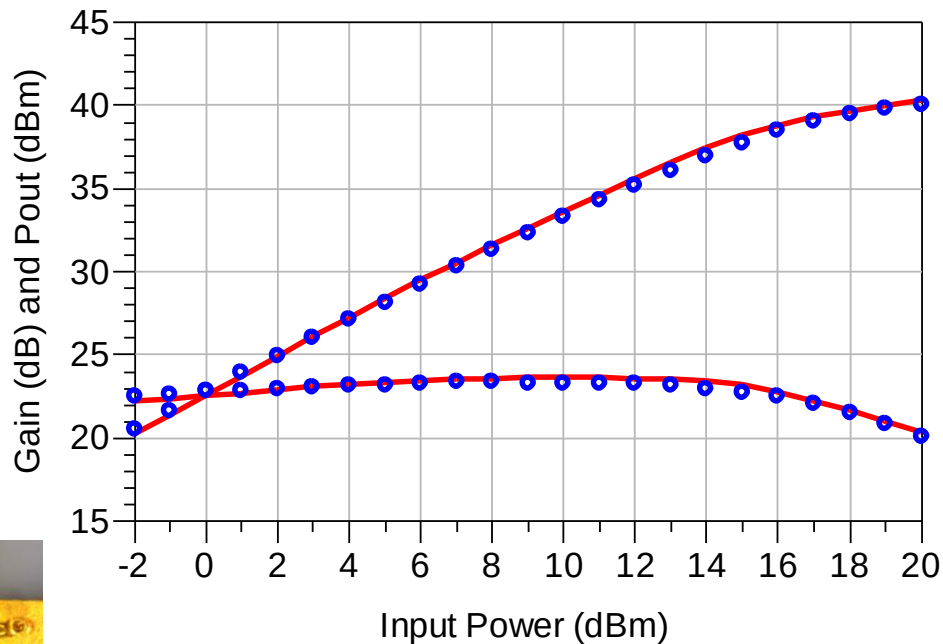
Legend: Red Solid lines – Model, Blue Solid lines – Measured.

Pulsed Biased conditions for measurements: duty cycle = 10% with pulse length = 100us.

Load Pull Summary	Max Power Load Reflection Coefficient	Max Power Value (dBm)	Max PAE Load Reflection Coefficient	Max PAE Value (%)
Measured	0.477<138°	39.9	0.582<101°	77.8
Model	0.448<142°	40.6	0.442<108°	74.5

Single Tone Power Sweep: Frequency = 3GHz
VDS = 28V, IDS = 50mA, 25C
Load Condition: Power Tuned

Transducer Gain and Output Power

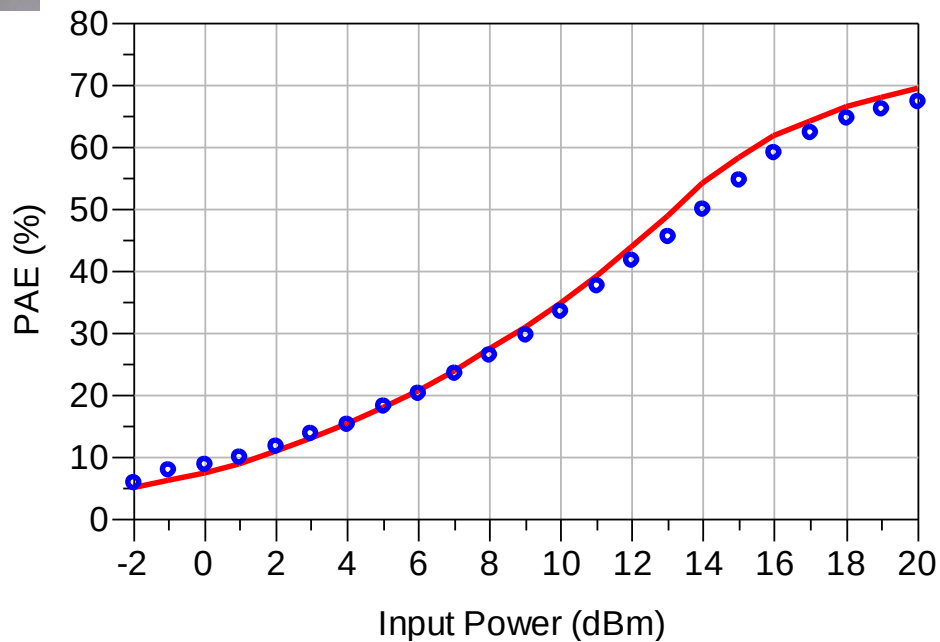


Model and Model and
Measurement Reference
Planes (BWremoval=0)

Test Bench Reflection
Coefficient:

GammaS=0.836<163°
GammaS2=0.472<75°
GammaS3=0.729<134°
GammaL=0.428<127°
GammaL2=0.449<-14°
GammaL3=0.508<-146°

Power Added Efficiency (PAE)



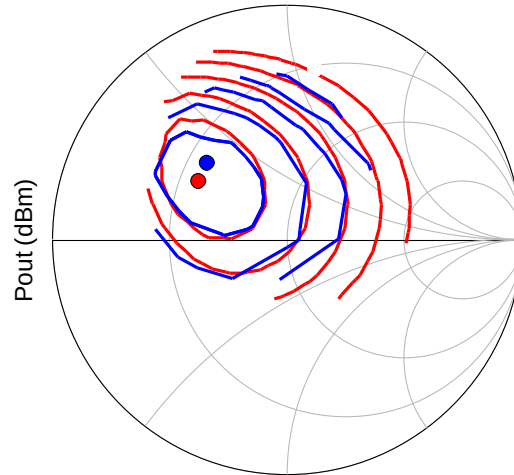
Legend: Red Solid lines - Model data, O Symbols - Measured data

Pulsed Biased conditions for measurements: duty cycle = 10% with pulse length = 100us.

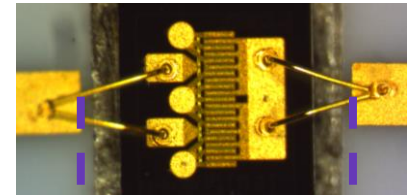
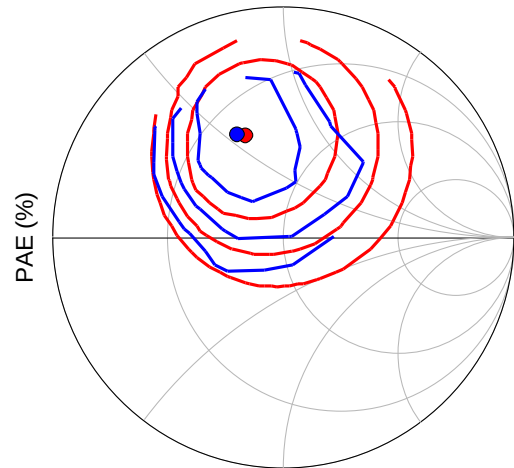
Load Pull Validation:

Frequency = 3GHz, VDS = 28V, IDS = 125mA,
Input Power = 19dBm, Z0 = 50Ω Center, 25C

Power Tuning (1dB contour step)



Efficiency Tuning (10% contour step)



Model and Model and
Measurement Reference
Planes (BWremoval=0)

Test Bench Reflection
Coefficient:

GammaS=0.857<163°
GammaS2=0.472<75°
GammaS3=0.729<134°
GammaL2=0.314<-32°
GammaL3=0.407<-170°

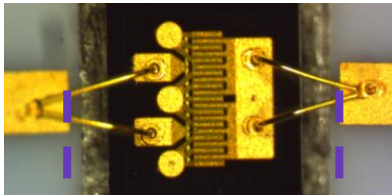
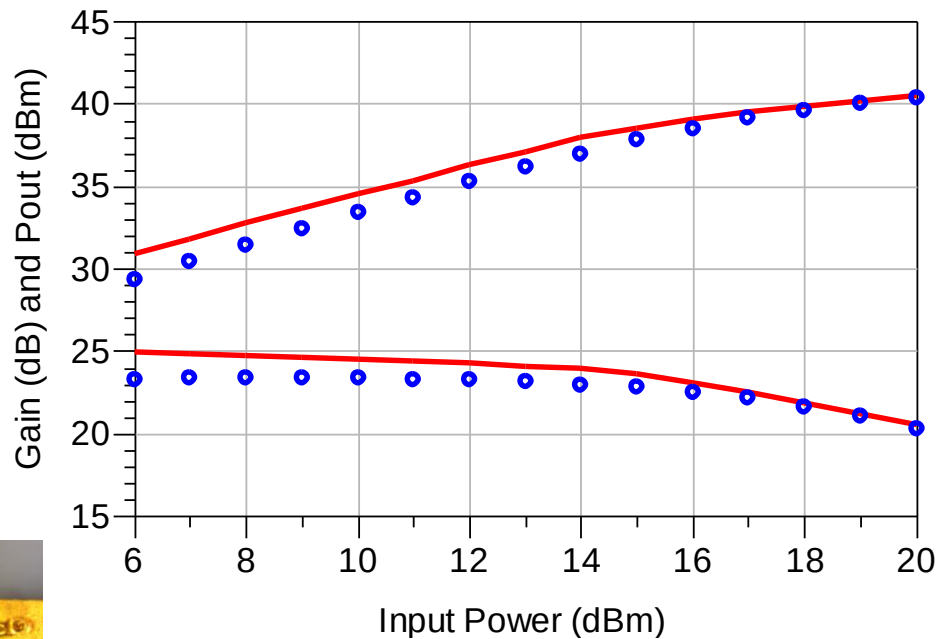
Legend: Red Solid lines – Model, Blue Solid lines – Measured.

Pulsed Biased conditions for measurements: duty cycle = 10% with pulse length = 100us.

Load Pull Summary	Max Power Load Reflection Coefficient	Max Power Value (dBm)	Max PAE Load Reflection Coefficient	Max PAE Value (%)
Measured	0.477<138°	40.3	0.480<111°	76.6
Model	0.455<147°	41.0	0.478<109°	75.0

Single Tone Power Sweep: Frequency = 3GHz
VDS = 28V, IDS = 125mA, 25C
Load Condition: Power Tuned

Transducer Gain and Output Power

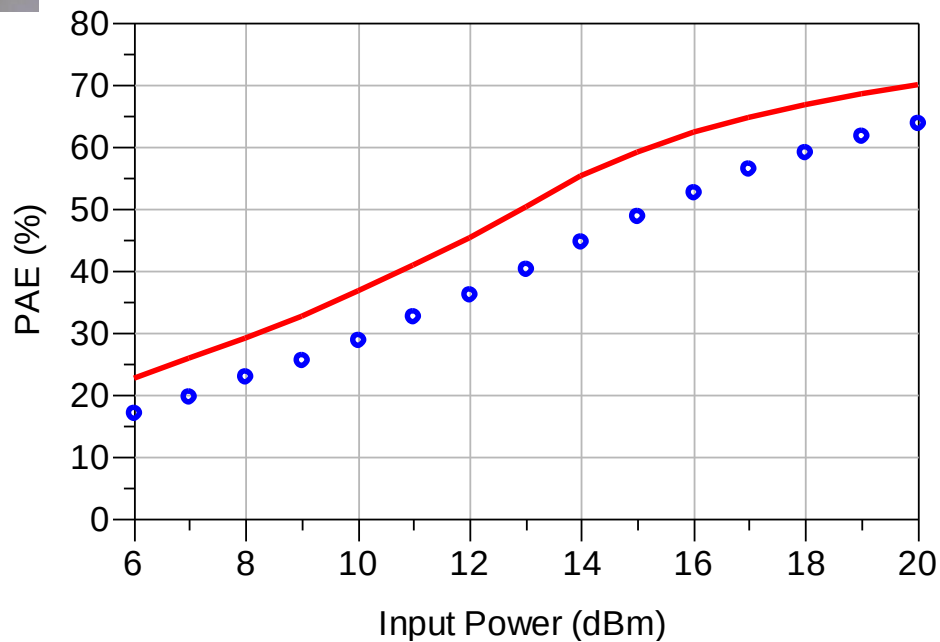


Model and Model and
Measurement Reference
Planes (BWremoval=0)

Test Bench Reflection
Coefficient:

GammaS=0.836<163°
GammaS2=0.472<75°
GammaS3=0.729<134°
GammaL=0.428<127°
GammaL2=0.449<-14°
GammaL3=0.508<-146°

Power Added Efficiency (PAE)



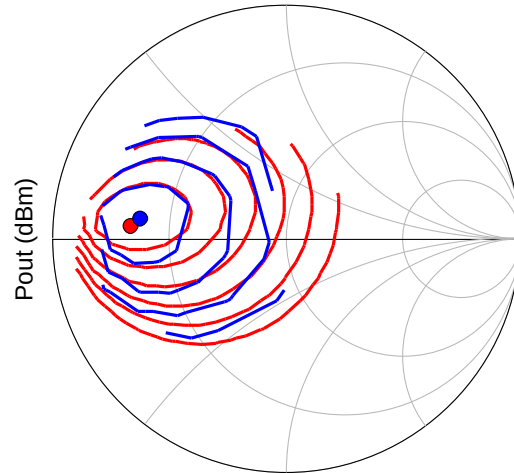
Legend: Red Solid lines - Model data, O Symbols - Measured data

Pulsed Biased conditions for measurements: duty cycle = 10% with pulse length = 100us.

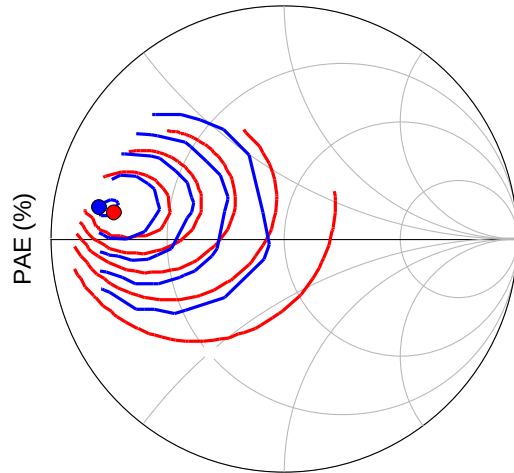
Load Pull Validation:

Frequency = 10GHz, VDS = 28V, IDS = 125mA,
Input Power = 30dBm, Z0 = 50Ω Center, 25C

Power Tuning (1dB contour step)

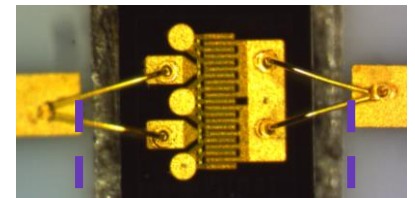


Efficiency Tuning (10% contour step)



Legend: Red Solid lines – Model, Blue Solid lines – Measured.

Pulsed Biased conditions for measurements: duty cycle = 10% with pulse length = 100us.



Model and Model and
Measurement Reference
Planes (BWremoval=0)

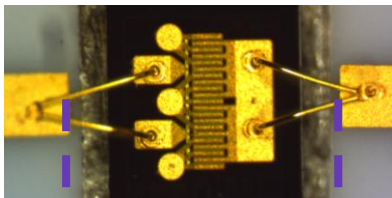
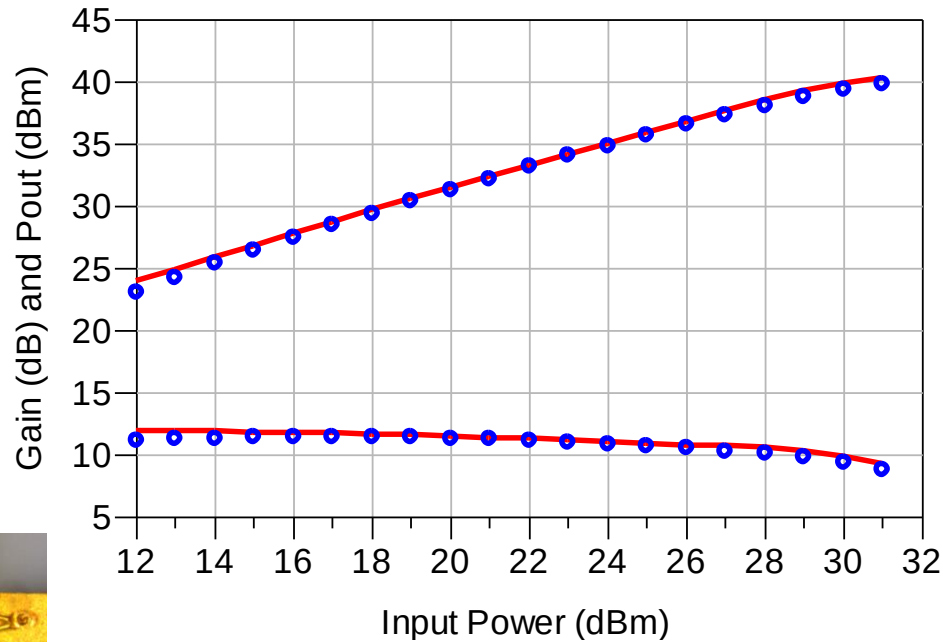
Test Bench Reflection
Coefficient:

GammaS=0.691<-152°
GammaS2=0.704<-161°
GammaS3=0.428<167°
GammaL2=0.577<-8°
GammaL3=0.306<-140°

Load Pull Summary	Max Power Load Reflection Coefficient	Max Power Value (dBm)	Max PAE Load Reflection Coefficient	Max PAE Value (%)
Measured	0.637<174°	39.5	0.802<171°	64.5
Model	0.666<176°	39.9	0.730<171°	59.4

Single Tone Power Sweep: Frequency = 10GHz
VDS = 28V, IDS = 125mA, 25C
Load Condition: Power Tuned

Transducer Gain and Output Power

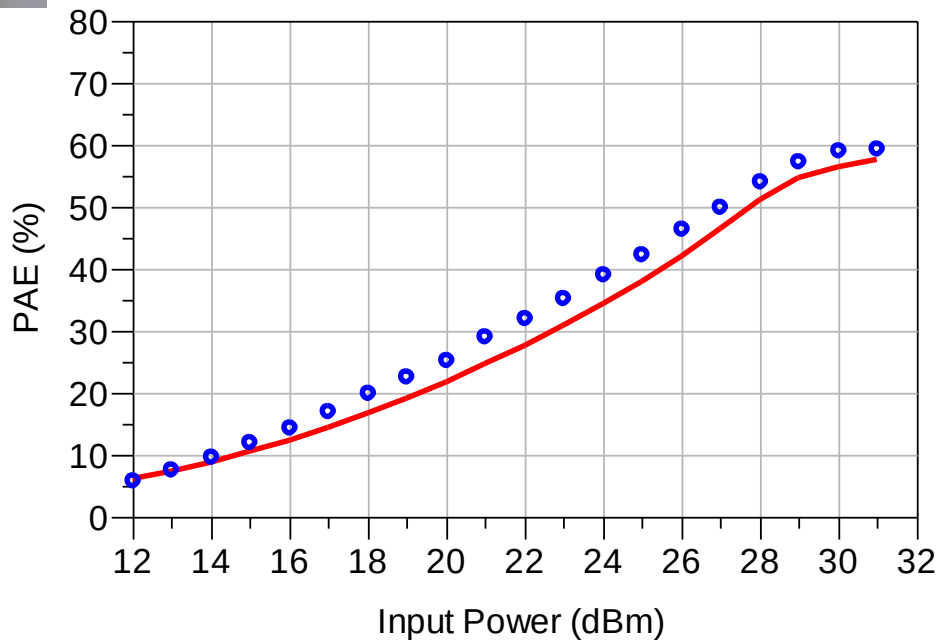


Model and Model and
Measurement Reference
Planes (BWremoval=0)

Test Bench Reflection
Coefficient:

GammaS=0.691<-152°
GammaS2=0.704<-161°
GammaS3=0.428<167°
GammaL=0.689<174°
GammaL2=0.684<107°
GammaL3=0.489<39°

Power Added Efficiency (PAE)

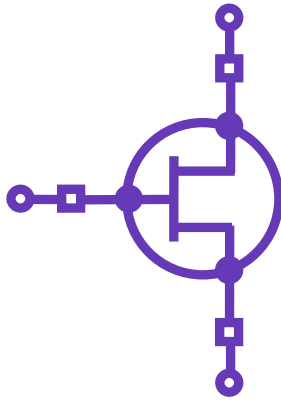


Legend: Red Solid lines - Model data, O Symbols - Measured data

Pulsed Biased conditions for measurements: duty cycle = 10% with pulse length = 100us.

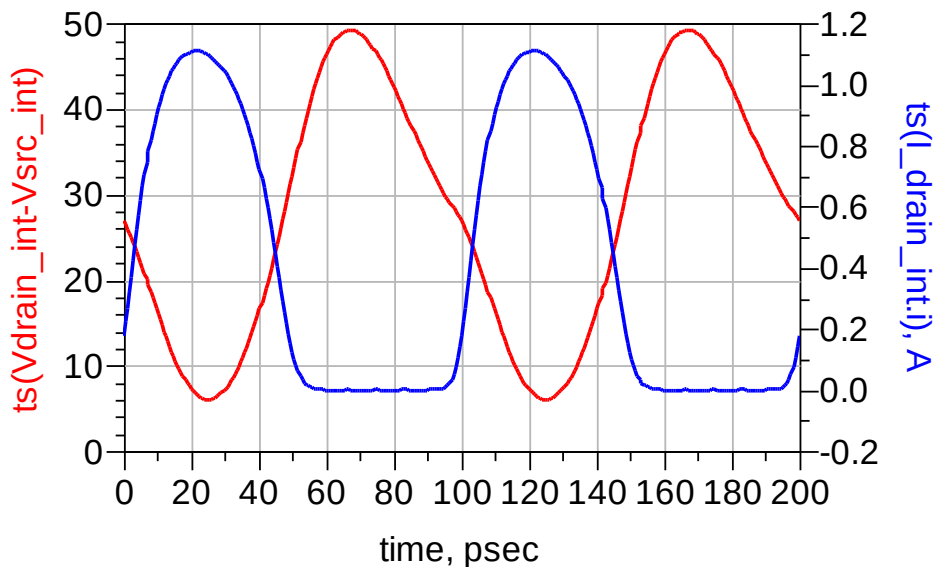
Advanced Model Features: Intrinsic Voltage/Current Sensing

Get V_{ds} and I_{ds} model data near current generator intrinsic planes while tuning.



- External Model Planes
- Internal Model Planes for I/V waveform analysis
- Parasitic networks available separately from intrinsic I/V model

Example Plot of internal node I_{ds} and V_{ds}



Results based on harmonic balance simulation at 28dBm input power, 10GHz, 28V, and 125mA.
 $\Gamma_{S1} = 0.691 \angle -152^\circ$, $\Gamma_{S2} = 0.704 \angle -161^\circ$, $\Gamma_{S3} = 0.428 \angle 167^\circ$, $\Gamma_{L1} = 0.689 \angle 174^\circ$, $\Gamma_{L2} = 0.684 \angle 107^\circ$, $\Gamma_{L3} = 0.489 \angle 39^\circ$.

Model and Datasheet Revision Notes

11/05/2014	Original model and datasheet development for HMT-TQT-TGF2023-2-02-001
01/14/2016	Model renamed and updated for improved 10 GHz efficiency. Added tuning parameters at inner layer for INTERNAL USE ONLY.